

Rithvik Nakirikanti

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Technical Skills

Languages: Java, Python, Go, JavaScript/TypeScript, SQL, Bash, Kotlin

Backend & Data: Spring Boot, Spring Batch, Dropwizard, PostgreSQL, Redis

Infrastructure: Docker, Kubernetes, Helm, OpenShift, Terraform, Argo CD, Jenkins

Interests: Backend Engineering, Distributed Systems, Cloud Infrastructure, Developer Tooling

Experience

Oracle Cloud Infrastructure | *Java, Dropwizard, OpenAPI, Kubernetes, Terraform*

Seattle, WA

Software Engineer 2

Jul 2025 – Present

- Launched PDB management features (create, rename) for Oracle Database@Google using **Java**, **Dropwizard**, and **OpenAPI** specs, introducing net-new capabilities to 500+ customers and reducing support tickets by 35%.
- Accelerated new region time-to-market for Oracle Database@Google by 75% (2 weeks → 3-4 days) by devising an automated infrastructure pipeline using **Terraform** to orchestrate **Kubernetes** cluster provisioning and control plane deployment, expanding availability across 8+ new regions for GCP customers.
- Designed and implemented scoped Resource Principal Session Tokens for the Oracle Database@Google control plane, enforcing least privilege via **JWT** claims and eliminating excessive access across 50+ customer database operations.
- Improved control plane resilience across 5+ customer workflows by instituting retry logic for intermittent downstream failures and strengthening **RESTful API** error handling, reducing workflow failures by 30%.

Verizon | *Java, Spring Boot, Spring Batch, Docker, Helm, Kubernetes, OpenShift, Jenkins, Argo CD*

Basking Ridge, NJ

Software Engineer Intern

Jun 2024 – Aug 2024

- Built a **Java Spring Boot** KPI ingestion microservice using **Spring Batch** to normalize 1,000+ network metrics from raw Verizon data sources into a canonical schema for downstream consumption.
- Designed **RESTful APIs** in **Spring Boot** to expose normalized KPI data to downstream dashboards, handling 500+ requests/minute at peak across 10+ internal teams with sub-200ms p95 latency.
- Optimized **Spring Batch** throughput via chunk-oriented processing and parallel step execution, reducing end-to-end data processing time by 5x and enabling same-day KPI refresh cycles.

Software Engineer Intern

Jun 2023 – Aug 2023

- Containerized PlatformZero services with **Docker/Helm** for **Kubernetes** deployment, then migrated releases from **Jenkins** to **Argo CD** under a **GitOps** model, eliminating configuration drift and improving deployment cadence by 50%.
- Refactored legacy configuration management in PlatformZero **Java** services to use **Spring** profiles, eliminating 50+ hardcoded environment values across 5+ services and simplifying deployments across dev, stage, and prod.
- Wrote **JUnit** and integration test suites achieving 85%+ code coverage across PlatformZero APIs, catching regressions pre-merge and reducing post-deployment hotfixes by 30%.

Avis Budget Group | *Kotlin, Android, Python, Tableau*

Parsippany, NJ

Software Engineer Intern

Jun 2022 – Aug 2022

- Developed an internal **Android** environment-switching tool in **Kotlin** that automated environment-specific configuration updates for 15+ QA engineers, eliminating manual setup steps and reducing testing overhead by 25%.
- Built a **Python** customer feedback pipeline using **BeautifulSoup** and **K-means clustering** to categorize 1,500+ reviews into actionable themes, surfacing insights in **Tableau** dashboards that informed service improvement initiatives.

Projects

Boston Public Library: Retrieval-Augmented Search | *Python, PostgreSQL, pgvector, OpenAI, Streamlit* **Fall 2024**

- Built a **RAG** pipeline over 1.2M+ Boston Public Library records using **PostgreSQL** with **pgvector**, enabling semantic search across 5M+ embedded text chunks with 2-5 second query latency.
- Designed chunking and embedding strategies for archival text, balancing chunk size and overlap to preserve context across paragraph boundaries and improve semantic match quality.
- Implemented **LLM**-based query expansion and **BM25** reranking using **OpenAI GPT-4o-mini**, improving retrieval accuracy for historical natural language queries over keyword-based search.

Education

Boston University

Boston, MA

Masters of Science in Computer Science & Bachelor of Arts in Computer Science

May 2025

• **GPA:** 3.80/4.00 (MS), 3.76/4.00 (BA)

• **Relevant Coursework:** Distributed Systems, Database Systems, Machine Learning, Computer Networks, Cloud Computing, Operating Systems

• **Leadership:** Teaching Assistant (Database Systems), Course Assistant (Comp. Systems)